CAFVIS: Hybrid Approach to Comprehensive Research Paper Summarization

Enhancing Factual Consistency and Comprehensiveness in Research Paper Summaries

Somnath Pandit
Dept. of Computer Science & Engg.
University Of Kalayni
Kalyani, India
somnathpandit633@gmail.com

Sayan Chandra
Dept. of Computer Science & Engg.
University Of Kalayni
Kalyani, India
sayanchandra89@gmail.com

Soumyajit Mukherjee
Dept. of Computer Science & Engg.
University Of Kalayni
Kalyani, India
sommajeet@gmail.com

Dr. Jyoti Sekhar Banerjee
Dept. of Computer Science & Engg.(AI & ML)
Bengal Institute of Technology
Kolkata, India
tojyoti2001@yahoo.co.in

***Abstract*—**This paper introduces CAFVIS (Context-Aware, Fact-Verified, and Interactive Summarizer), a novel hybrid framework designed to address key limitations in current text summarization systems—particularly their inability to process embedded visual content and ensure factual accuracy. CAFVIS integrates state-of-the-art components in Natural Language Processing (NLP), Computer Vision, and multimodal AI to generate coherent, verifiable and context-rich summaries from complex research documents.

The architecture combines abstractive and extractive summarization, factual verification, and visual-textual fusion through a robust pipeline that leverages tools such as GROBID, LXML, Pegasus, PixTral, BLIP-2, RoBERTa-MNLI, SBERT and Command-R+. By bridging the gap between textual understanding and visual information extraction, CAFVIS significantly reduces hallucinations and improves factual alignment in summaries. Its design supports query-focused retrieval, making it highly applicable to research-intensive and enterprise-level environments.

***Index Terms*—Hybrid AI, Document Understanding, Multimodal AI, Abstractive Summarization, Factual Verification, Research Paper Analysis, Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP), Computer Vision.**

I. Introduction

THE proliferation of scientific and technical literature, especially in PDF format, presents a crucial challenge for efficient information extraction and summarization. Traditional text-based processing systems often fall short in summarizing the full scope of a research paper, particularly when very important information is embedded within visual contents such as figures, graphs, charts and diagrams. To solve this problem, the CAFVIS (Context-Aware, Fact-Verified, Interactive Summarizer) architecture proposes a sophisticated hybrid methodology that seamlessly integrates both textual and visual data streams. This system is primarily designed to produce high-quality, contextually relevant and factual summaries from intricate textual sources. The hybrid approach within CAFVIS signifies a collaborative combination of diverse, state-of-the-art artificial intelligence and machine learning techniques. This encompasses advanced Natural Language Processing (NLP) for abstractive summarization and factual verification, cutting-edge Computer Vision and multimodal AI for image and document understanding and robust document parsing technologies. It is a hybrid summarization system that combines both textual and visual information to generate meaningful summaries of very long research papers. It is structured into three key phases: Document Storage, Abstract Summary Generation and Extractive Summary Generation. In the first phase, tools like GROBID, PixTral, BLIP-2 and Command-R+ are used to parse and store structured data in a vector database for RAG-based processing. The abstractive summary is generated using Pegasus and verified for accuracy with RoBERTa-MNLI and factual snippets. In the final phase, SBERT and MMR are used to filter out top-K ranked sentences from the original document with a similarity with the query for a query-focused extractive summary. This paper provides a detailed analysis of the CAFVIS architecture, elaborating each major component, its technical specifications, and its contribution to the system's overarching goal of delivering verifiable and comprehensive insights from research papers.

II. Literature Review

1) Abstractive Text Summarization: The proposed text summarization system is designed to generate concise summaries while preserving the most critical information from large input texts. This system employs an abstractive summarization technique utilizing a deep learning-based Long Short-Term

Memory (LSTM) sequence-to-sequence model. Thereby facilitating quicker and more efficient information retrieval Anitha et al [1].

2) PEGASUS for Abstractive Summarization: Pegasus is a transformer based encoder-decoder model tailored for abstractive summarization, introducing a **gap sentence** generation pre-training objective that masks and predicts the important sentences from the document, closely mimicking summarization. For further information come to Jingqing et al [2].

3) EXABSUM: This research introduces EXABSUM, a novel automatic text summarization approach that generates both extractive and abstractive summaries where the extractive component select key sentences and the abstractive component rewrites content in a more concise, coherent form. For further observation and information, come to Zakariae et al [3].

4) Transformer-Based Summarization: This research presents various text summarization models and presented the results of it's own implementation of automatic text summarizer which was trained using the CNN Daily mail dataset. The Transformer-based model effectively improves summary accuracy over time with training. The transformer is a deep learning framework that attempts to address sequential processes while addressing long-range persistent data with minimal difficulty. For more information, go through Shruti et al [4].

5) Z-Code++: It proposes a new encoder-decoder PLM optimized for abstractive summarization, Z-Code++, which significantly extends Z-Code a state-of-the-art PLM developed for machine translation. The model extends the state of the art encoder-decoder model. Here a two-phase pre-training process is used to improve model's performance on low-resource summarization tasks and the self-attention layers are replaced in the encoder with disentangled attention layers. For further reading and information retrival, come to Pengcheng et al [5].

6) Bridging Extractive and Abstractive Summarization: This proposed summarizer introduces GenX, a novel approach that reframes extractive summarization as an inference task handled by abstractive (generative) models, thereby unifying the two traditionally distinct paradigms. Instead of relying on oracle-labeled sentences or customized extractive architectures, GenX leverages pre-trained sequence-to-sequence models to perform extractive summarization. Notably, GenX preserves its ability to generate abstractive summaries, making it a flexible dual-purpose solution. For full and detailed information retrival and research Daniel et al [6].

7) Memory-Efficient Long Document Summarization: It introduces LOCOST, a novel encoder-decoder architecture that leverages state-space models (SSMs) instead of self-attention to perform abstractive summariza- tion of extremely long documents. Unlike the transformer based models which face scalability issues LOCOST achieves a significantly lower complexity of O(L log L). The results highlight LOCOST's promise as a lightweight, scalable alternative to transformer-based architectures for long-context natural language processing tasks. For more information and observation reach at Florian et al [7].

8) Chapter-wise PDF Summarization: This research presents a novel AI-based summarization model aimed at easing academic workloads for students by generating concise chapter-wise summaries from textbooks or PDF documents. The system leverages Google's T5 model to convert entire pages of a document into readable summaries using minimal computational resources, making it feasible for low-spec devices. For further information retrival, come to Anita et al [8].

9) Multimodal Vision-Language Models for Document Understanding: Recent advancements in multimodel learning have led to the invention of models like PixTral and BLIP-2, which play a crucial role in processing visual contents like scientific graphs, plots, bar chart, diagram, image etc. PixTral [9] (Pixel-Level Transformer for Layout-Aware Document Understanding) uses a Vision encoder for image processing and Mistral-12B multimodel decoder to give textual representations of the input images. On the other hand BLIP-2 [10] (Bootstrapped Language Image Pretraining v2) represents a significant advancement in vision-language pretraining. This model basically introduces two stages training approach that decouples the vision encoder from the language model, allowing it to efficiently utilize large-scale pretrained LLMs like OPT and T5. Together, PixTral and BLIP-2 provide a robust foundation for multimodal document processing.

III. Research Gaps of the existing AI models

A. Contextual Limitations

Research papers are often long, sometimes exceeding 10,000 words or more. But most AI models (such as BERT or GPT) can only read a limited number of words at once (called "token limits").

B. Factual Inconsistency (Hallucinations)

When models don't have full context, they often fill the gaps with educated guesses, which may be wrong. So in science and research, wrong information can mislead readers or create false understanding.

C. Multimodal Understanding

Research papers contain images, graphs, tables, equations, references, and jargon. AI models are mostly trained on plain text. Current summarization systems offer limited user control or verification, making it hard to trust or interact with the generated output. They can't properly process the visual contents in a research paper.

D. Transparency and Control

Users cannot see how the summaries are generated or customize them. This lack of transparency makes it difficult for users to understand or influence the summarization process, which can lead to mistrust in the system and less effective use of AI-generated summaries in critical research contexts.

E. Limits in Selective Search and Filtering

Most summarizers can't target specific sections or queries; T5 allows only manual page picks without dynamic filtering. CAFVIS adds query-based search and section-level filtering for precise, scalable summaries.

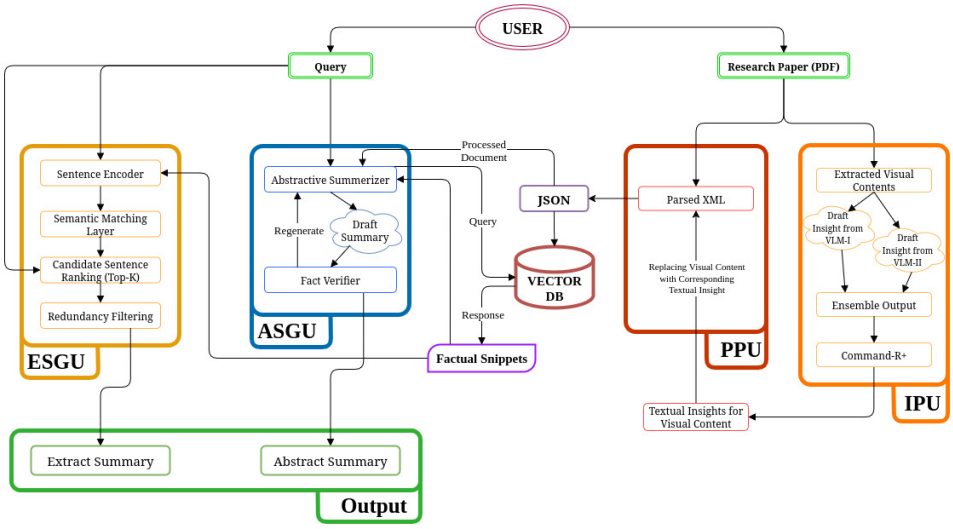

Figure 1: Proposed CAFVIS model architecture

IV. Methodology

The CAFVIS data flow architecture is designed as a multi-stage pipeline, as depicted in the provided diagram, ensuring a thorough and integrated processing of both textual and visual information.

As shown in Fig. 1, the CAFVIS pipeline integrates both text and visual processing modules.

A. Input Layer: User Query and Document Ingestion

The process is initiated by two primary inputs: one is "User Query" and the other is "Research Paper" especially in PDF format. This initial integration suggests that the user's specific information need or question guides the subsequent processing, enabling query-biased summarization or question-answering over document.

B. Document Parsing and Structuring

1) PDF Parsing Unit: The Research Paper first enters the GROBID [11] module, which is a machine learning library, designed using java specifically for extracting, parsing, and re-structuring raw documents, particularly PDFs, into structured TEI-XML encoded documents, with a strong focus on technical and scientific publications. Its functionalities include header extraction, reference parsing, and full text structuring, encompassing paragraphs, section titles, figures, and tables. Another image extraction tool, such as pdf2image + LayoutParser, is used to extract visual regions out of the PDF to use them in the Image Processing Unit later.

PDF Parsing Unit (PPU)
A. Utilities: GROBID, LXML python library
B. Input: PDF
C. Output: Structured & Enriched JSON
D. Role: Convert the PDF into structured text to store in Vector Database.

2) Image Processing Unit: The pages of the PDF is converted to images using pdf2image and for the extraction of the visual regions LayoutParser is used to crop them out. Now the cropped and extracted images are fed to two separate Vision Language Models (VLMs), PixTral [9] and BLIP-2 [10]. The outputs, which describe the visual content in text format, are combined in a single prompt. In the next step, the combined prompt is provided to Command-R+ to generate a refined, comprehensive and insightful interpretation of the given image. Finally, the image is replaced with its interpretation and a final text-enriched XML content is formed, without any loss of information.

Image Processing Unit (IPU)
A. Utilities: pdf2image, LayoutParser, PixTral, BLIP-2, Command-R+
B. Input: PDF
C. Output: Textual Insight for the Visual Content
D. Role: Convert a visual content into it's textual description.

The XML is converted to a JSON content using python's LXML library.

C. Final Textual Insight Generation

The Ensemble Output (representing a pre-defined and combined prompt containing of the textual interpretations from the visual content processing pipeline) is directly fed into Command-R+ [12] to produce the final textual insight. Command-R+ is Cohere's powerful, scalable Large Language Model (LLM) built for enterprise use cases. It features a substantial 128k-token context window, critical for integrating and synthesizing large volumes of information. This model is used here to normalize the textual representations, given by PixTral and BLIP-2 models, and to produce a final structured textual representation of the visual contents. Its key capabilities

include advanced Retrieval-Augmented-Generation (RAG) [13] with citation to reduce hallucinations and tool use to automate sophisticated business processes and also visual question answering. Command-R+ also supports multi-step tool use and self-correction, enhancing its ability to handle complex reasoning tasks.

D. Factual Grounding

The processed document from the PDF Parsing Unit (in JSON format), is embedded and stored into a vector database to perform Retrieval-Augmented-Generation (RAG). On receiving a query, "Factual Snippets" are generated as response from the vector database using RAG. These "Factual Snippets" are stored temporarily allowing for semantic search and grounding of language model outputs to mitigate hallucination.

E. Abstract Summary Generation Unit

1) Abstract Summary Generation: The "User Query" is directly fed into the "Abstract Summary Generator Unit" (ASGU, and particularly to Pegasus) to produce a "Draft Summary". Pegasus is an encoder-decoder transformer model pretrained on unlabeled text to perform abstractive summarization. Its training involves Masked Language Modeling (MLM) and Gap Sentence Generation (GSG). The google/pegasus-arxiv model, trained on academic datasets (C4 and HugeNews) and with a tokenizer updated to encode newline characters, is particularly well-suited for abstractive summarization of structured documents like research papers.

2) Factual Verification: The "Draft Summary", generated by Pegasus, and the "Factual Snippets", retrieved from the vector database, are both fed into the "Fact Verifier" (RoBERTa-MNLI). The output of this verification step is the "Active Output" (Abstract Summary). This is a crucial step for ensuring the factual accuracy and consistency of the generated summary. RoBERTa-MNLI is a Transformer-based language model fine-tuned on the Multi-genre Natural Language Inference (MNLI) corpus. Its core task is to predict the relationship between a "premise" sentence and a "hypothesis" sentence, namely whether the premise entails the hypothesis, contradicts it, or is neutral. This capability allows the verifier to check if statements in the "Draft Summary" are factually supported by or contradict the "Factual Snippets".

Abstract Summary Generator Unit (ASGU)
A. Utilities: Pegasus, RoBERTa-MNLI
B. Input: Query, Snippets, Complete Processed Content
C. Output: Abstract Summary
D. Role: Provides an abstract overview on the topic asked by user.

F. Extract Summary Generation

Parallel to the ASGU, "User Query" is directly fed to the "Extract Summary Generation Unit" (ESGU) too. This unit comprises of four layers:

1) Sentence Encoder: The "User Query" along with the "Factual Snippets" is particularly fed to a sentence encoder (SBERT) [14] in ESGU. SBERT encodes the "User Query" and the "Factual Snippets" and forwards them to the next stage which is "Semantic Matching Layer".

2) Semantic Matching Layer: Each sentence of the "Factual Snippets" are ranked based on the similarity (cosine) found between the encoded query and the sentences. Next, this layer forwards a ranked list of sentences to the next layer.

3) Top-K Candidate Sentence Ranking: Out of the ranked list of encoded sentences, this layer filters out first K sentences from the list, sorted in descending order by their ranking. Selected sentences are next forwarded to the next layer.

4) Redundancy Filtering: In this layer, redundancy is removed from the selected sentences to make the extract summary more compact and straightforward. This layer uses MMR (Maximal Marginal Relevance) algorithm to filter out redundant words/sentences. Now the final extract summary is generated as Passive Output.

Extract Summary Generator Unit (ESGU)
A. Utilities: SBERT, MMR (Algorithm)
B. Input: Query, Snippets, Complete Processed Content
C. Output: Textual Insight for the Visual Content
D. Role: Filters out Top-K sentences from the original document related to user query.

TABLE I: Comparison of Key Features in Popular Summarization Models

Model	Factual Accuracy	Multimodal Understanding	Interactivity & Output Flexibility	Long-Document Handling	Selective Search
LSTM [1]	Low; drifts	None	No control; abstractive	Short texts only	No
EXABSUM [3]	High extractive; mid abstractive	Text	None; hybrid	Short inputs only	No
Z-Code++ [5]	High; few errors	Text	None; abstractive	Long docs via segmentation	No
GenX [6]	Mid; drifts	Text	None; hybrid	Medium; weak on long	No
LOCOST [7]	Good; human-checked	Text	None; abstractive	Very long, efficient	No
T5 [8]	Variable	None	Page-based; abstractive	Manual page-by-page	Limited
CAFVIS (Proposed)	Verified; source-checked	Text + images, tables, graphs	Full control; custom style	Very long; targeted	Yes; query, filter, section summaries

V. Theoretical Analysis of Summarization Models

As shown in Table I, summarization models differ greatly in how they are designed, what they can do, and what they aim to achieve. Early sequence models like LSTM are quick, but often drift from the facts and work only with text. Hybrid systems such as EXABSUM improve fact retention through a mix of extractive and abstractive techniques, but still don't allow users to interact with the results. Transformer-based approaches like Z-Code++ and GenX handle context better, yet remain text-only and lack user-driven content focus. LOCOST can process very long documents efficiently, while T5 offers basic page-by-page control—but still needs manual section selection and does not support true multimodal input.

CAFVIS takes these capabilities further by combining NLP and computer vision tools (Pegasus, GROBID, PixTral, BLIP-2, Command-R+), making it possible to process both text and visual elements such as tables, charts, and graphs in one seamless workflow. It delivers fact-checked, interactive, and query-focused summaries, and importantly, adds selective search and filtering so users can pinpoint specific sections or topics in long academic documents. This approach fills long-standing gaps in factual reliability, multimodal understanding, and customizable output—positioning CAFVIS as a forward-looking, researcher-friendly model for the future of summarization.

VI. Discussion and Limitations

CAFVIS effectively combines extractive and abstractive summarization with factual verification and semantic grounding, ensuring that generated summaries are both informative and reliable. Its integration of Pegasus, GROBID, Vector DB, and RoBERTa-MNLI provides a strong pipeline for handling structured academic content. However, the system faces limitations such as domain dependency performance may drop on non-academic or niche texts and increased processing time due to multiple sequential components. The quality of factual grounding also depends heavily on the accuracy of snippet extraction and vector matching. Additionally, while BERTScore aids in semantic evaluation, it does not measure factual correctness, making human oversight important in critical applications.

VII. Conclusion

CAFVIS is a unified, hybrid system built to deliver reliable, multimodal summaries of academic documents addressing the long-standing shortcomings of text-only, non-interactive tools. Uniquely, it allows users to "chat" with a paper, making the summarization process more intuitive and interactive. By combining advanced PDF parsing (GROBID), visual text fusion (PixTral, BLIP-2), hybrid summarization (Pegasus, SBERT), and factual verification (RoBERTa-MNLI, RAG), CAFVIS produces summaries that are accurate, context-rich, and tailored to specific user needs.

It can manage long, complex research papers; incorporate visuals such as figures and tables; and verify details directly against the source. Its query-driven, interactive design ensures both transparency and adaptability. In doing so, CAFVIS sets a benchmark for future summarization frameworks—where factual accuracy, multimodal integration and user control are central to the design.

References

[1] Dr.J.Anitha, M.Raahavi, and S. M.Rehapriadarsini, "Abstractive text summarization," *Journal of Xidian University*, vol. 14, no. 6, 2020, https://doi.org/10.37896/jxu14.6/094. [Online]. Available: http://xadzkjdx.cn/

[2] J. Zhang, Y. Zhao, M. Saleh, and P. J. Liu, "Pegasus: Pre-training with extracted gap-sentences for abstractive summarization," *Proceedings of the 37th International Conference on Machine Learning*, vol. 119, pp. 11 328–11 339, 2020, https://proceedings.mlr.press/v119/zhang20ae.

[3] Z. A. Merrouni1, B. Frikh1, and B. Ouhbi, "Exabsum: a new text summarization approach for generating extractive and abstractive summaries," *Journal of Big Data*, vol. 10, 24 October, 2023, https://link.springer.com/article/10.1186/s40537-023-00836-y?

[4] S. J. S. T. A. S. Kapse and M. Atique, "Analysis and implemented on automated text summarization using transformer model," *International Journal of Engineering and Advanced Technolo*, April 2024, https://www.researchgate.net/publication/380207947_Analysis_and_Implemented_on_Automated_Text_Summarization_using_Transformer_Model.

[5] P. He1, B. Peng, S. Wang, Y. Liu, R. Xu, H. H. Awadalla, Y. S. C. Zhu, W. Xiong, M. Zeng, J. Gao, and X. Huang, "Z-code++: A pre-trained language model optimized for abstractive summarization," *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, vol. 1, July, 2023, https://aclanthology.org/2023.acl-long.279/.

[6] D. Varab and Y. Xu, "Abstractive summarizers are excellent extractive summarizers," *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, vol. 2, July, 2023, https://aclanthology.org/2023.acl-short.29/.

[7] F. L. Bronnec, S. Duong, M. Ravaut, A. Allauzen, N. F. Chen, V. Guigue, A. Lumbreras, L. Soulier, and P. Gallinari, "Locost: State-space models for long document abstractive summarization," *Cornell University*, vol. 1, 31st Jan, 2024, https://arxiv.org/abs/2401.17919.

[8] Anita, B. Mathew, and S. Kurian, "Novel artificial intelligence model for chapter- wise summarization from books," *Viswajyothi College of Engineering and Technology*, vol. 12, Jun, 2024, https://www.ijraset.com/research-paper/novel-artificial-intelligence-model-for-chapter-wise-summarization-from-books.

[9] P. Agrawal, S. Antoniak, and E. B. Hanna, "Pixtral 12b: A 12-billion-parameter multimodal language model," *arXiv preprint arXiv:2410.07073*, vol. v2, Oct 2024, https://doi.org/10.48550/arXiv.2410.07073.

[10] J. Li, D. Li, S. Savarese, and S. Hoi, "Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," *arXiv preprint arXiv:2301.12597*, vol. v3, Jun 2023, https://doi.org/10.48550/arXiv.2301.12597.

[11] P. Lopez, "Grobid: Combining automatic bibliographic data recognition and term extraction for scholarship publications," *arXiv preprint arXiv:2003.12218*, vol. v1, March 2020, https://arxiv.org/abs/2003.12218.

[12] C. Labs, "Command r+:a 104-billion-parameter open-weights model for retrieval-augmented generation and tool-use," *Cohere Research / arXiv preprint*, vol. v1, Apr 2024, https://openlaboratory.ai/models/command-r-plus.

[13] Z. P. Lee, A. Lin, and C. Tan, "Finetune-rag: Fine-tuning language models to resist hallucination in retrieval-augmented generation," *arXiv preprint arXiv:2505.10792*, vol. v2, May 2025, https://arxiv.org/abs/2505.10792.

[14] M. Mokoatle, V. Marivate, D. Mapiye, R. Bornman, and V. M. Hayes, "A review and comparative study of cancer detection using machine learning: Sbert and simcse application," *BMC Bioinformatics*, vol. 24, no. 112, March 2023, https://doi.org/10.1186/s12859-023-05235-x.